

A

STM Exercise 3

- Creep
- Fatigue

CREEP

creep test: a constant load is applied at a given T , strain is recorded versus time.

Constant load in the elastic field can give to a plastic deformation, progressive to fracture.

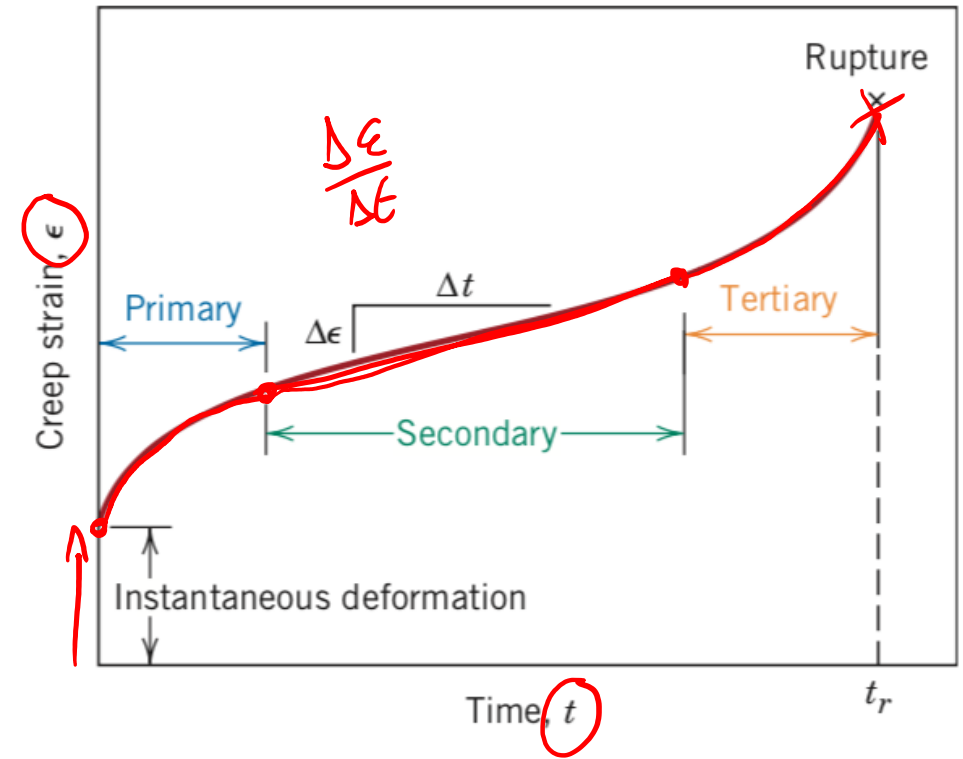
1. Thermally activated process (High T Process)

Metals $T > 0,3-0,4 T_{\text{melting}}$

Ceramics $T > 0,6-0,7 T_{\text{melting}}$

Amorphous materials $T > T_{\text{glass}}$

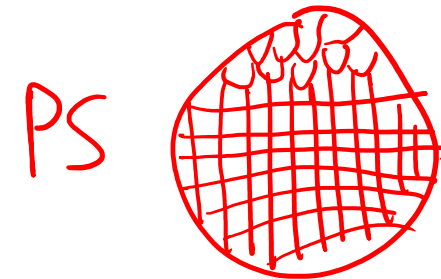
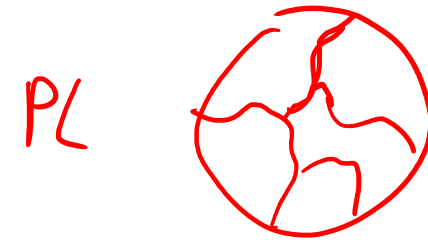
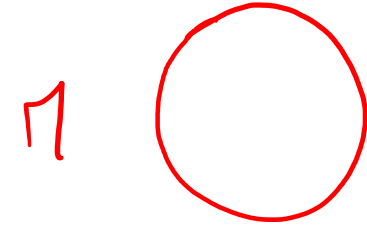
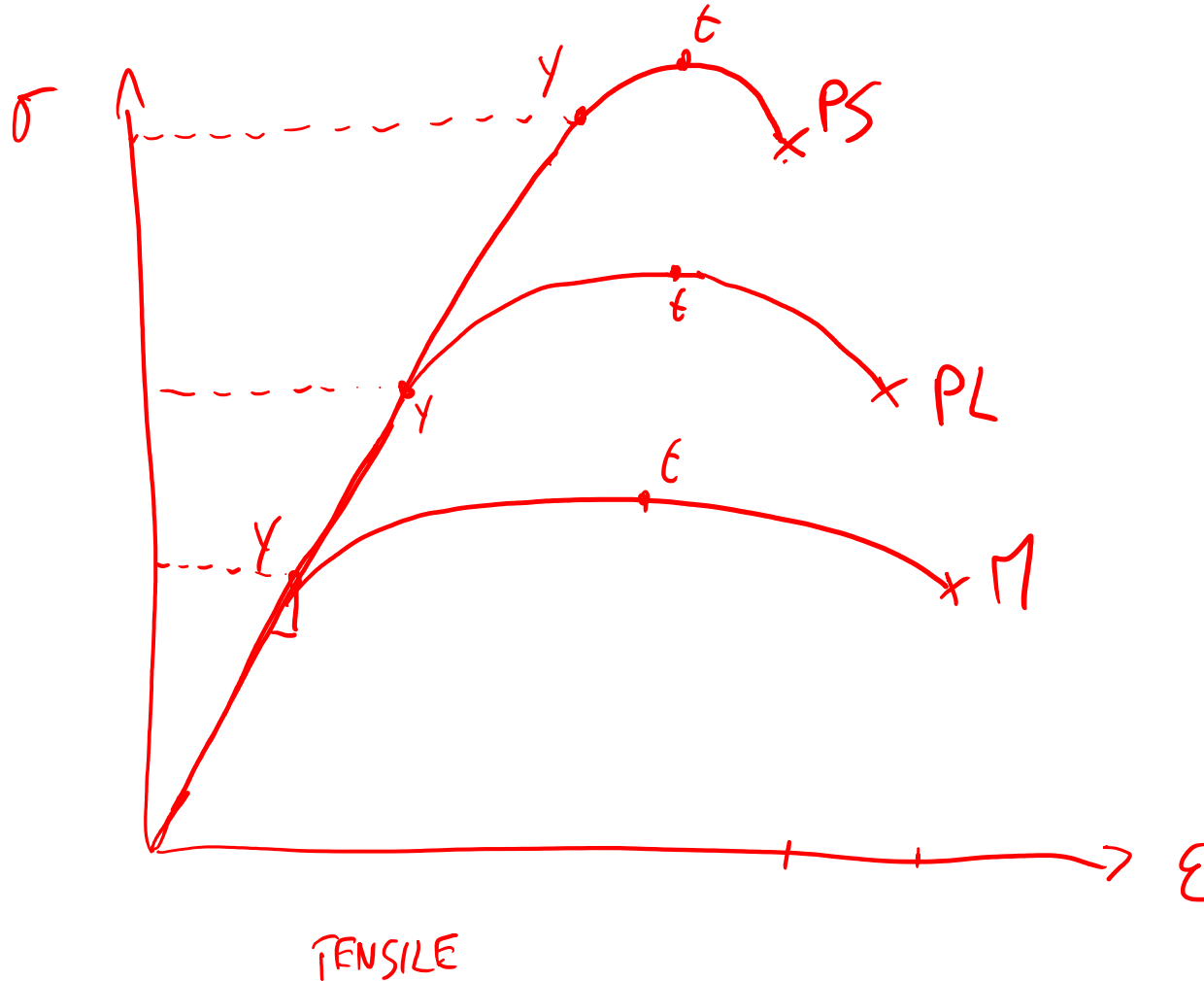
$\sigma = \text{const}$



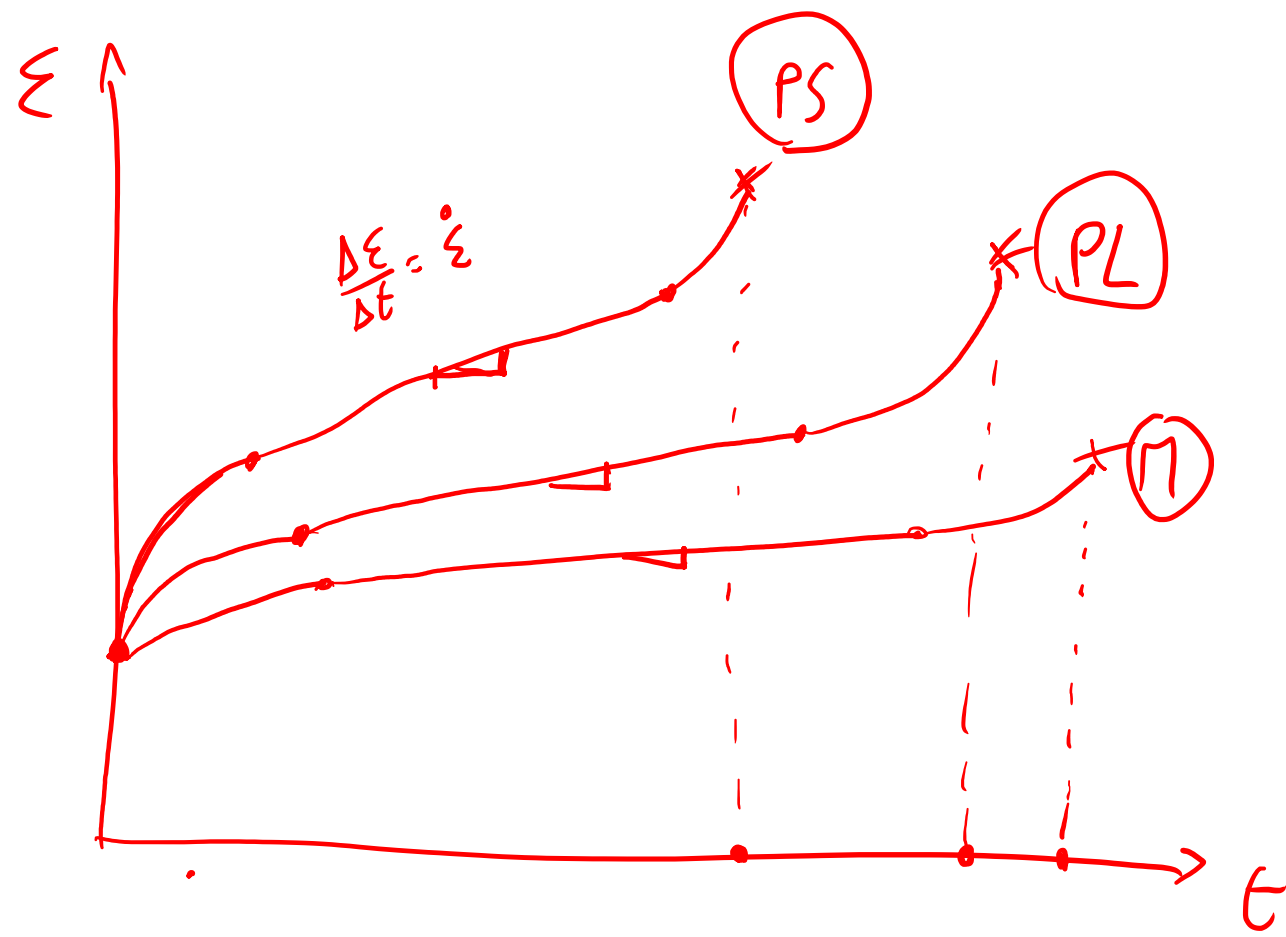
1- Draw tensile and creep curves of a given metal with the following microstructures and Briefly explain the differences: mono-crystalline, polycrystalline with large grains, polycrystalline with small grains.

1- Draw tensile and creep curves of a given metal with the following microstructures and Briefly explain the differences: mono-crystalline, polycrystalline with large grains, polycrystalline with small grains.

η PL PS

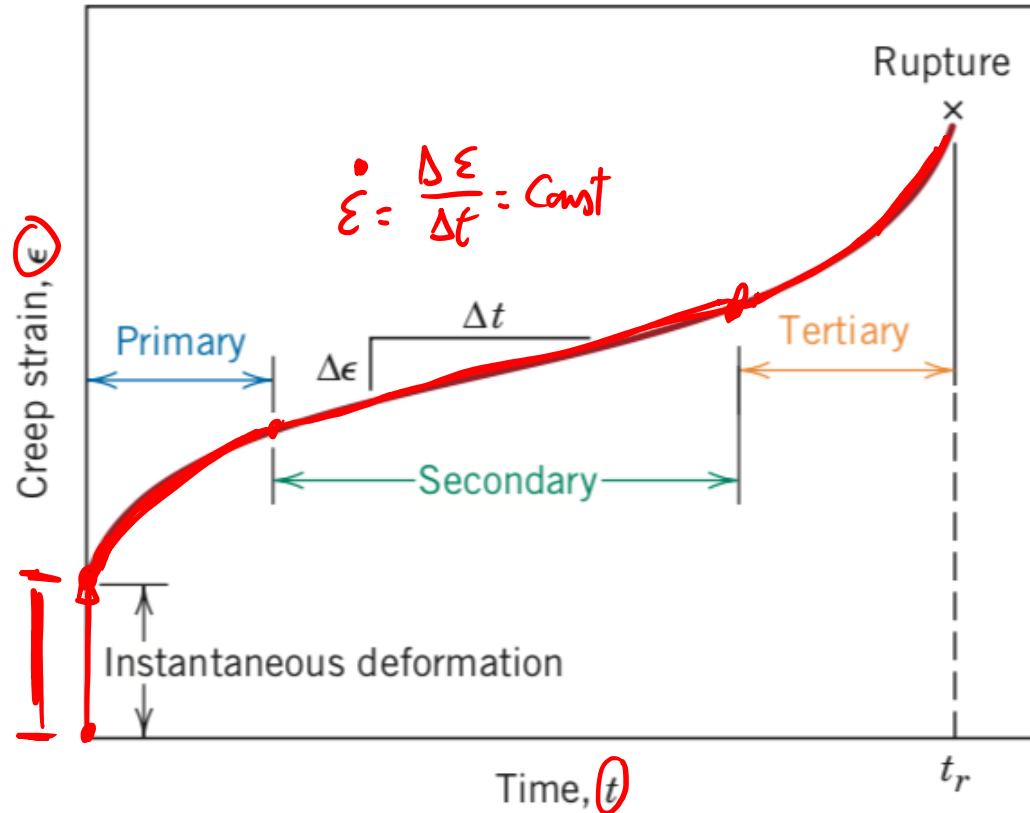


1- Draw tensile and creep curves of a given metal with the following microstructures and Briefly explain the differences: mono-crystalline, polycrystalline with large grains, polycrystalline with small grains.



2- Draw the creep curve of materials and briefly discuss the three phases of creep in term of materials micro-structural modification. Draw the creep curves of the same material at $\sigma_1 > \sigma_2$ and at $T_1 > T_2$. Explain differences.

2- Draw the creep curve of materials and briefly discuss the three phases of creep in term of materials micro-structural modification. Draw the creep curves of the same material at $\sigma_1 > \sigma_2$ and at $T_1 > T_2$. Explain differences.



CREEP I:

cold working higher than annealing.

CREEP II:

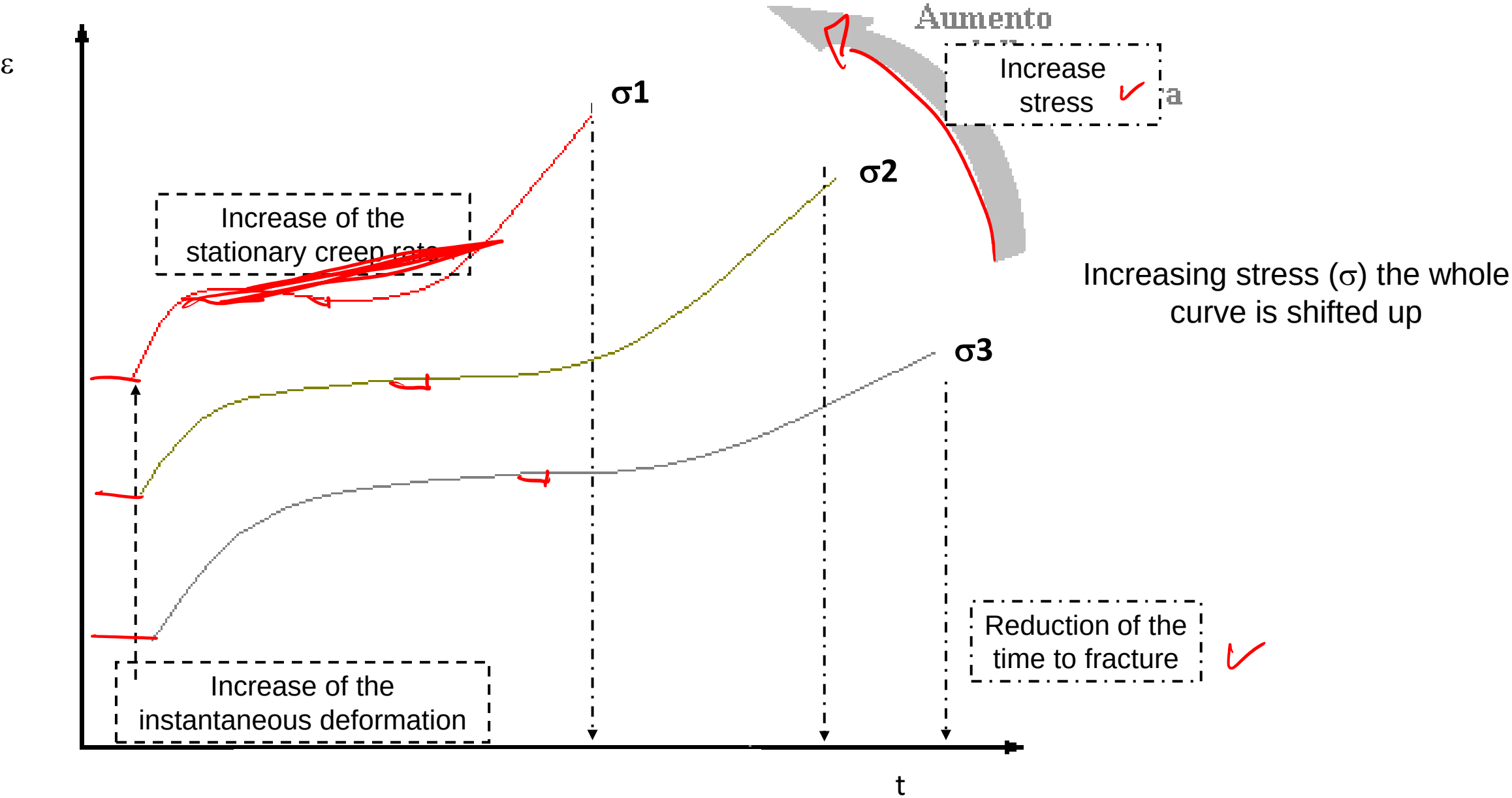
constant strain: balance between cold working and annealing

CREEP III:

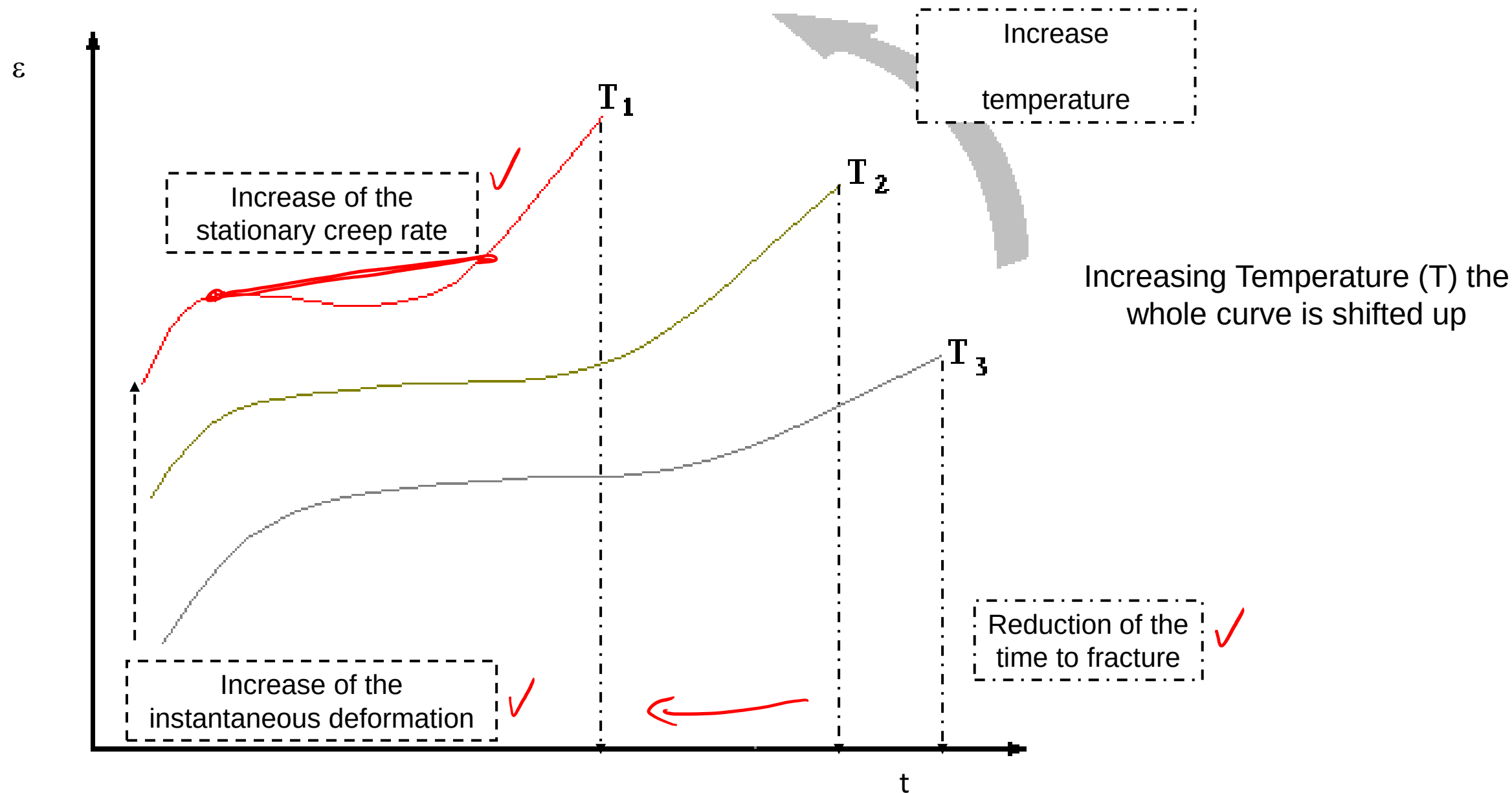
micro-cavities and other macro-defects at the grain boundaries: fracture of the sample

2- Draw the creep curve of materials and briefly discuss the three phases of creep in term of materials micro-structural modification. Draw the creep curves of the same material at $\sigma_1 > \sigma_2$ and at $T_1 > T_2$. Explain differences.

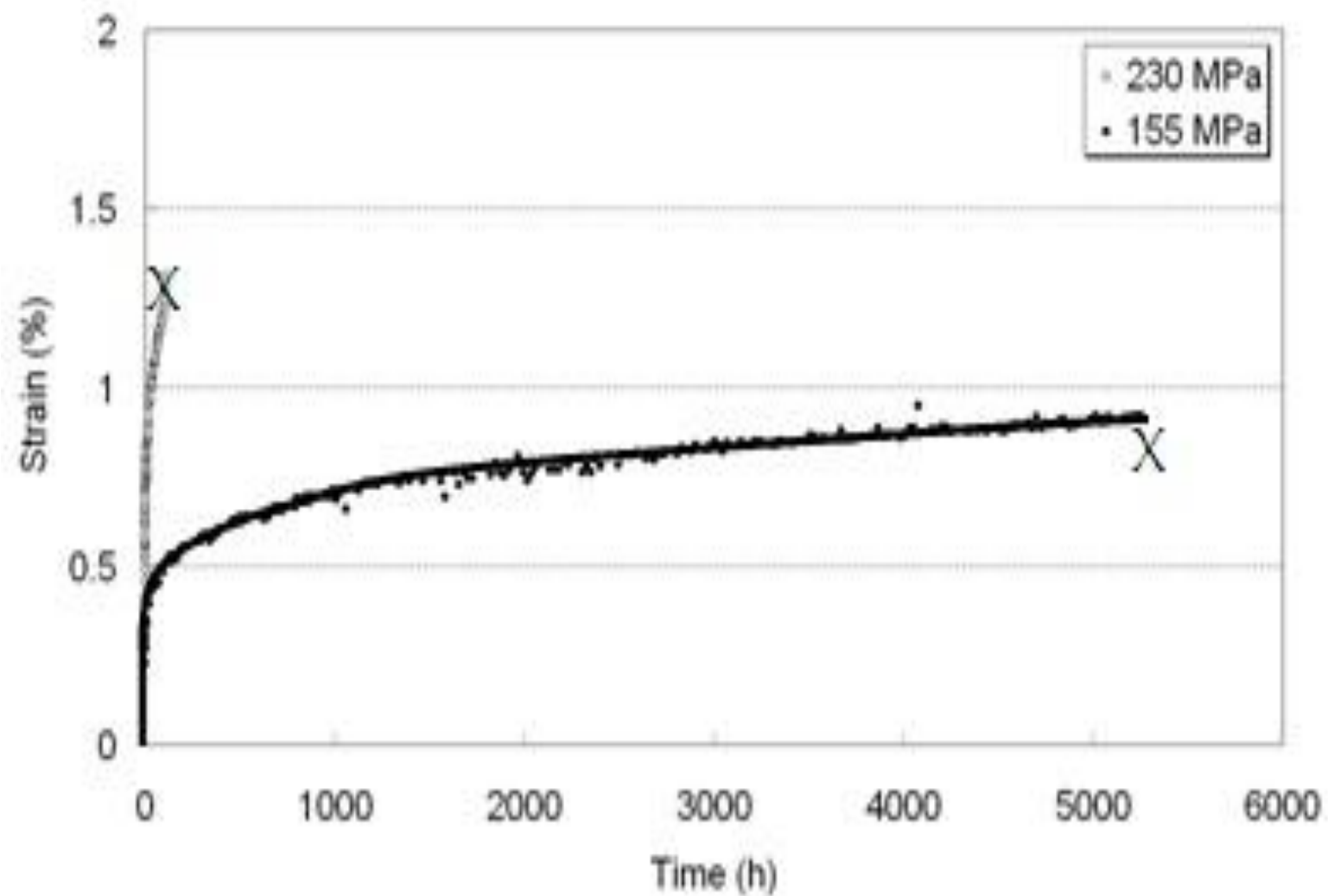
2- Draw the creep curve of materials and briefly discuss the three phases of creep in term of materials micro-structural modification. Draw the creep curves of the same material at $\sigma_1 > \sigma_2$ and at $T_1 > T_2$. Explain differences.

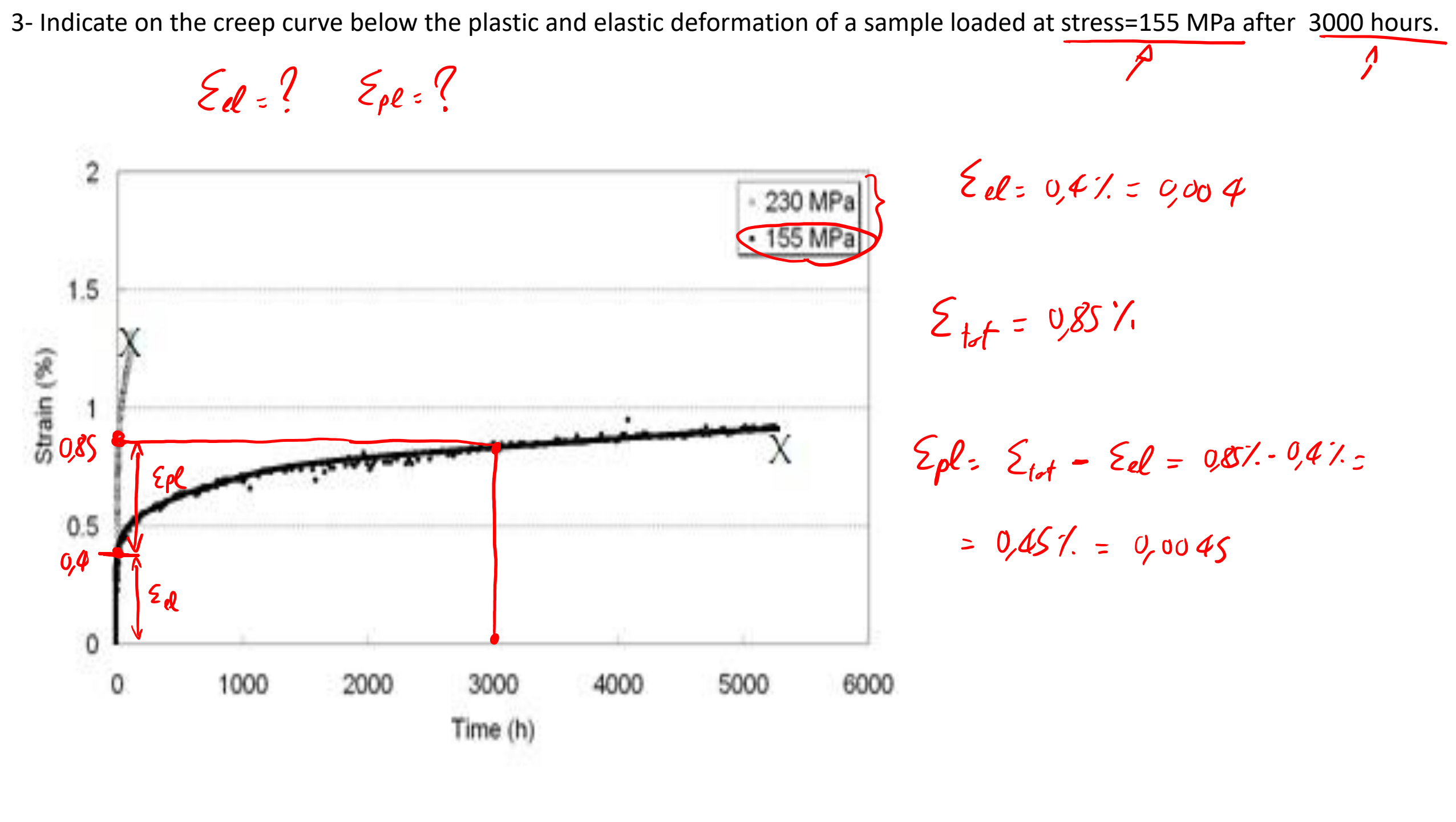


2- Draw the creep curve of materials and briefly discuss the three phases of creep in term of materials micro-structural modification. Draw the creep curves of the same material at $\sigma_1 > \sigma_2$ and at $T_1 > T_2$. Explain differences.



3- Indicate on the creep curve below the plastic and elastic deformation of a sample loaded at stress=155 MPa after 3000 hours.

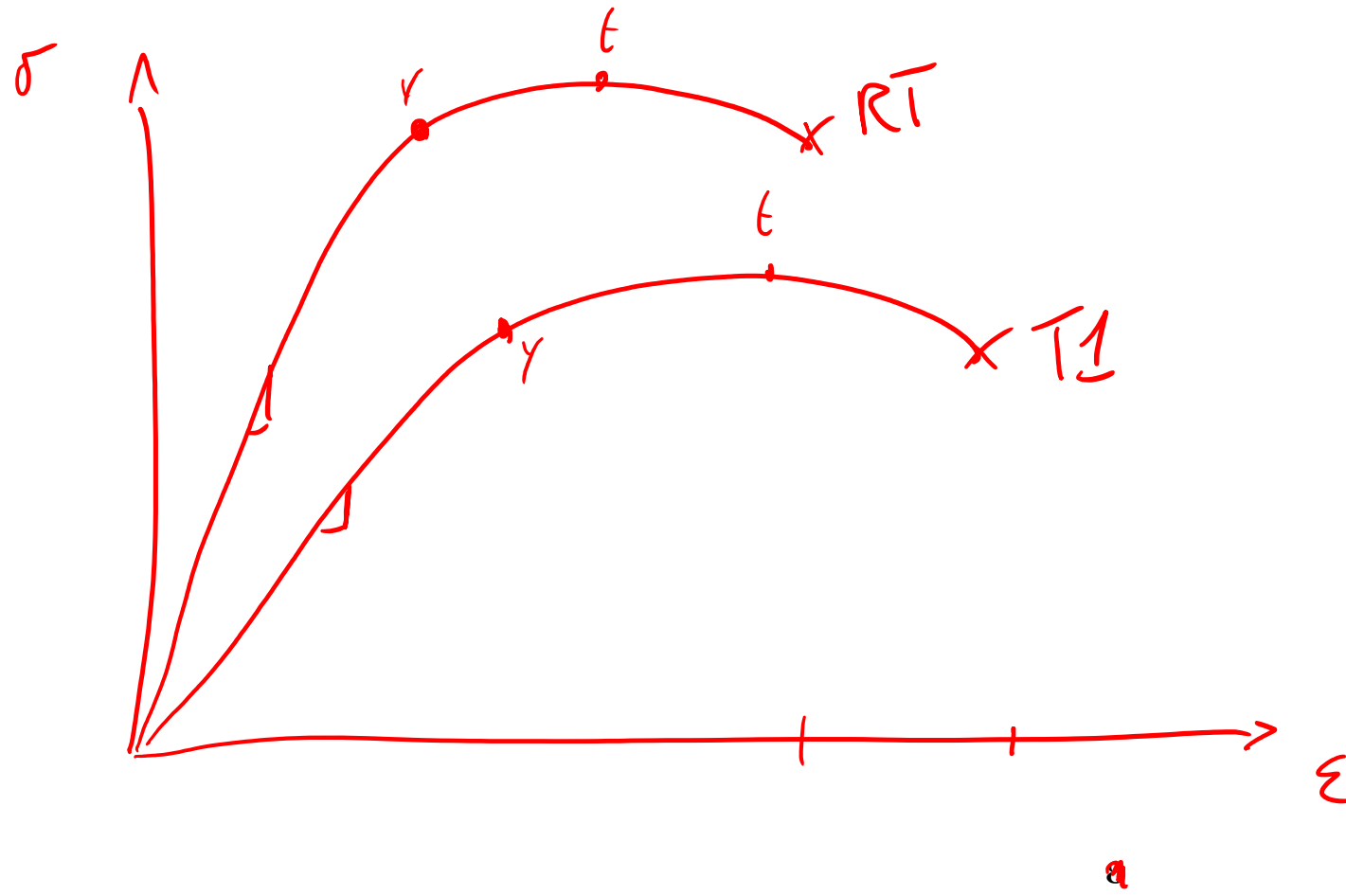




4- Draw the stress/strain curves after tensile tests of a metal at room temperature (RT) and at $T_1 > RT$. Briefly explain the difference.

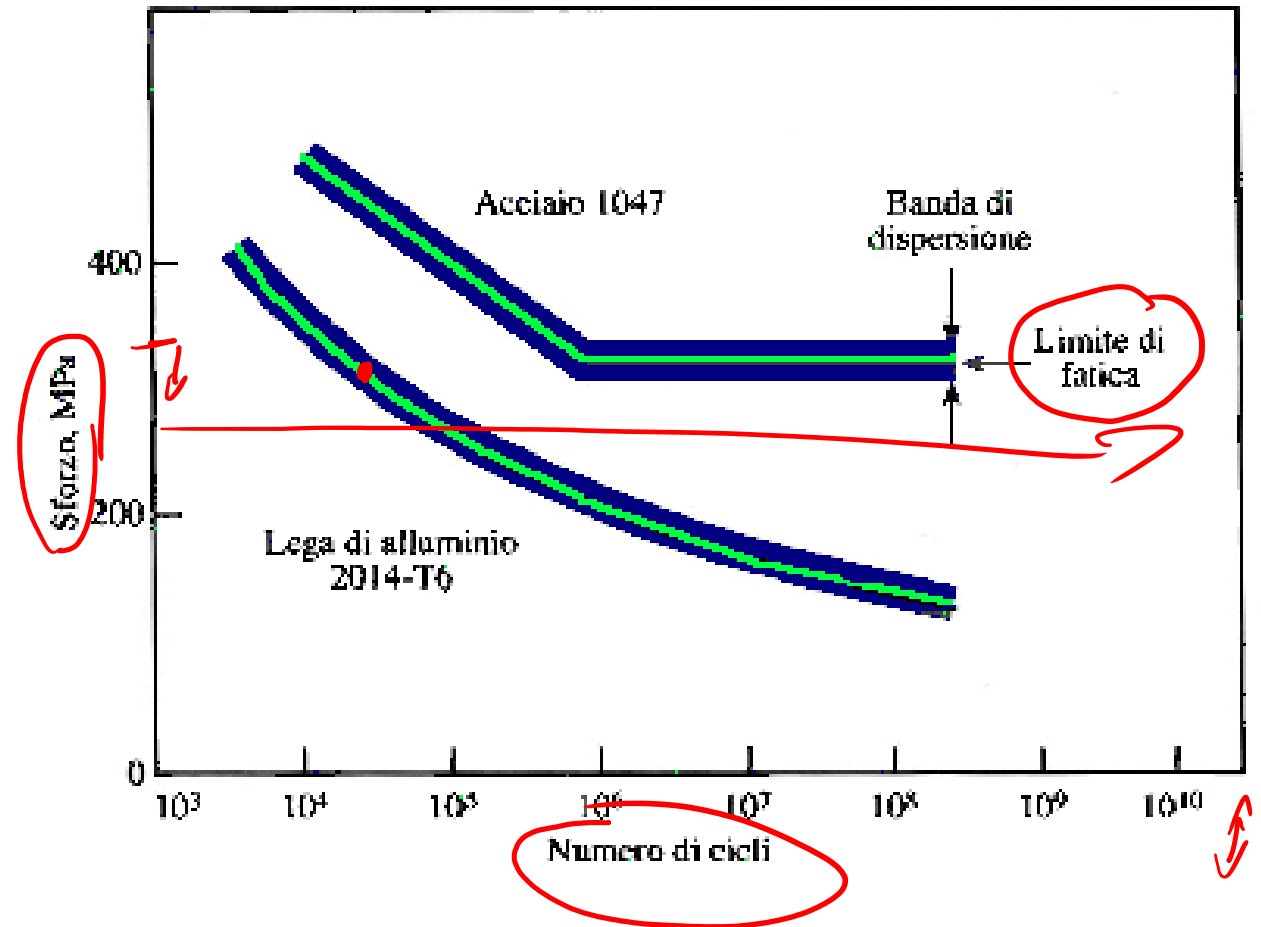
ϵ

4- Draw the stress/strain curves after tensile tests of a metal at room temperature (RT) and at $T_1 > RT$. Briefly explain the difference.



Fatigue

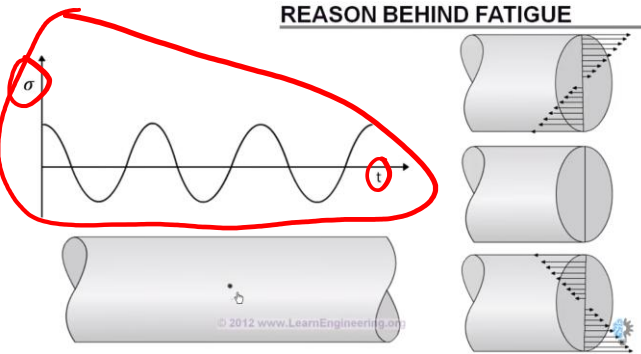
- Failure of materials due to cyclic loading.
- Main reason of mechanical failure of materials
- Failure happens at stress lower than σ_R or σ_Y
- Catastrophic failure of materials (also for ductile materials !)
- Fatigue tests: materials are cyclically loaded at different stresses up to failure.
- Fatigue limit: when cyclically loaded below this limit, materials do not fail (time considered by the graph)



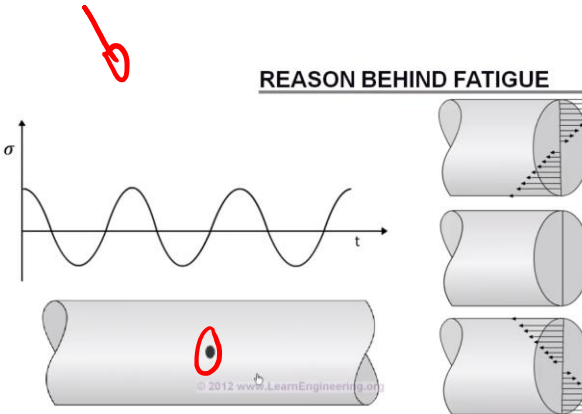
Fatigue

More cycles

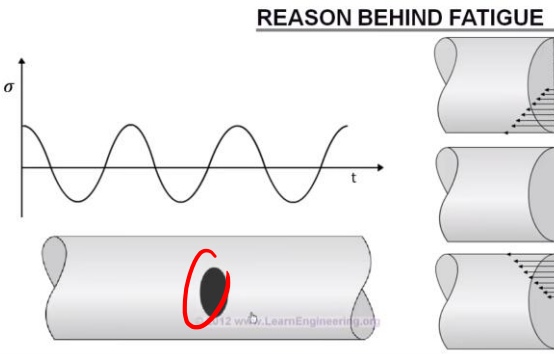
REASON BEHIND FATIGUE



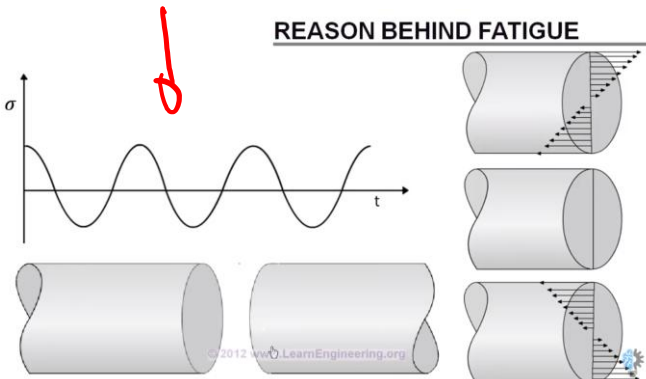
REASON BEHIND FATIGUE



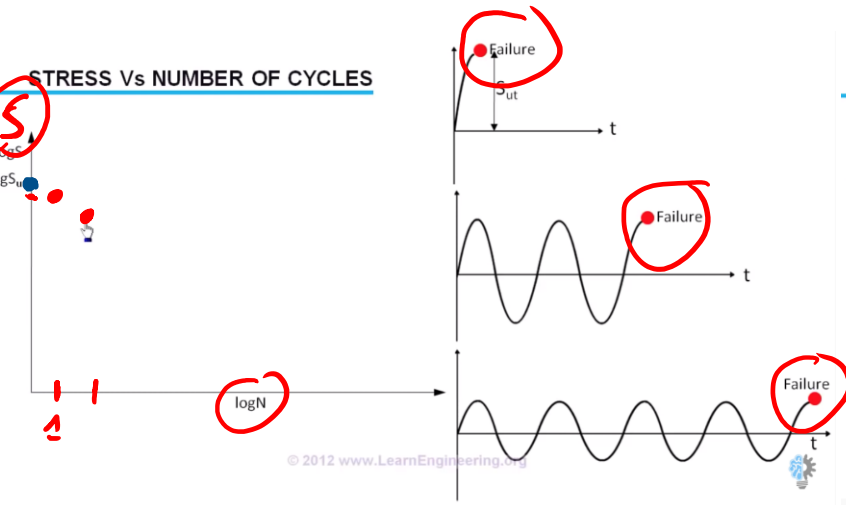
REASON BEHIND FATIGUE



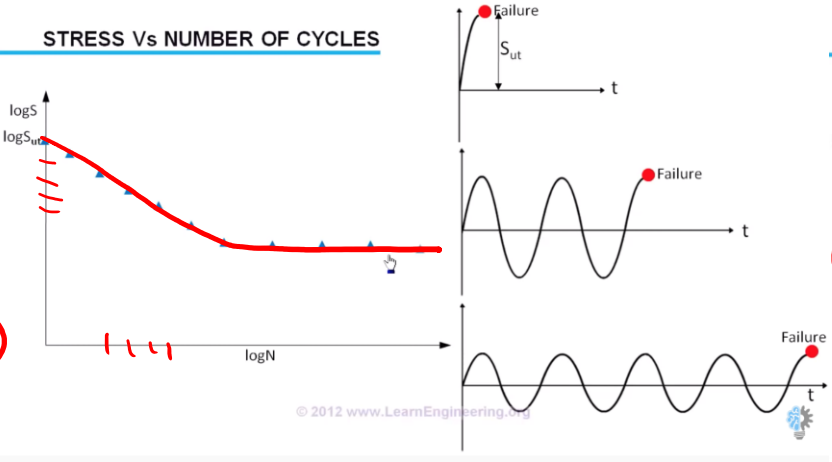
REASON BEHIND FATIGUE



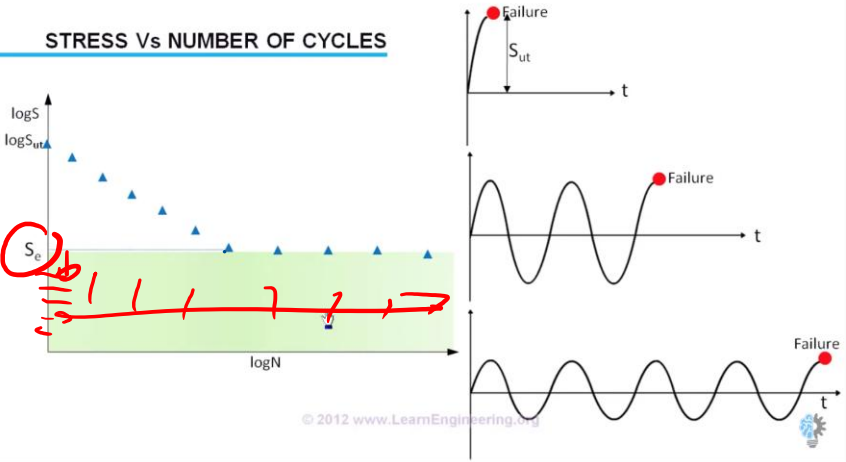
STRESS Vs NUMBER OF CYCLES



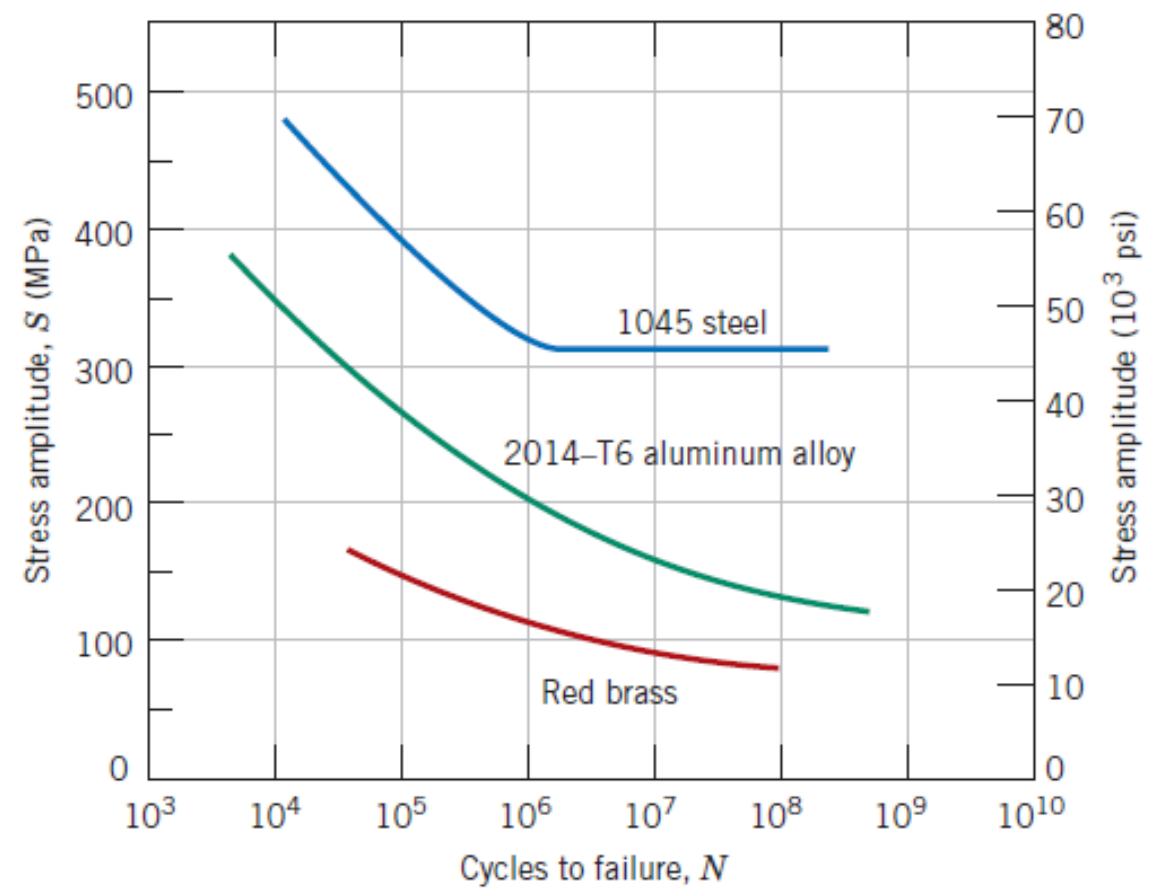
STRESS Vs NUMBER OF CYCLES



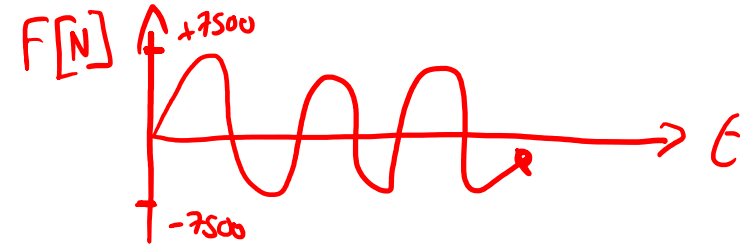
STRESS Vs NUMBER OF CYCLES



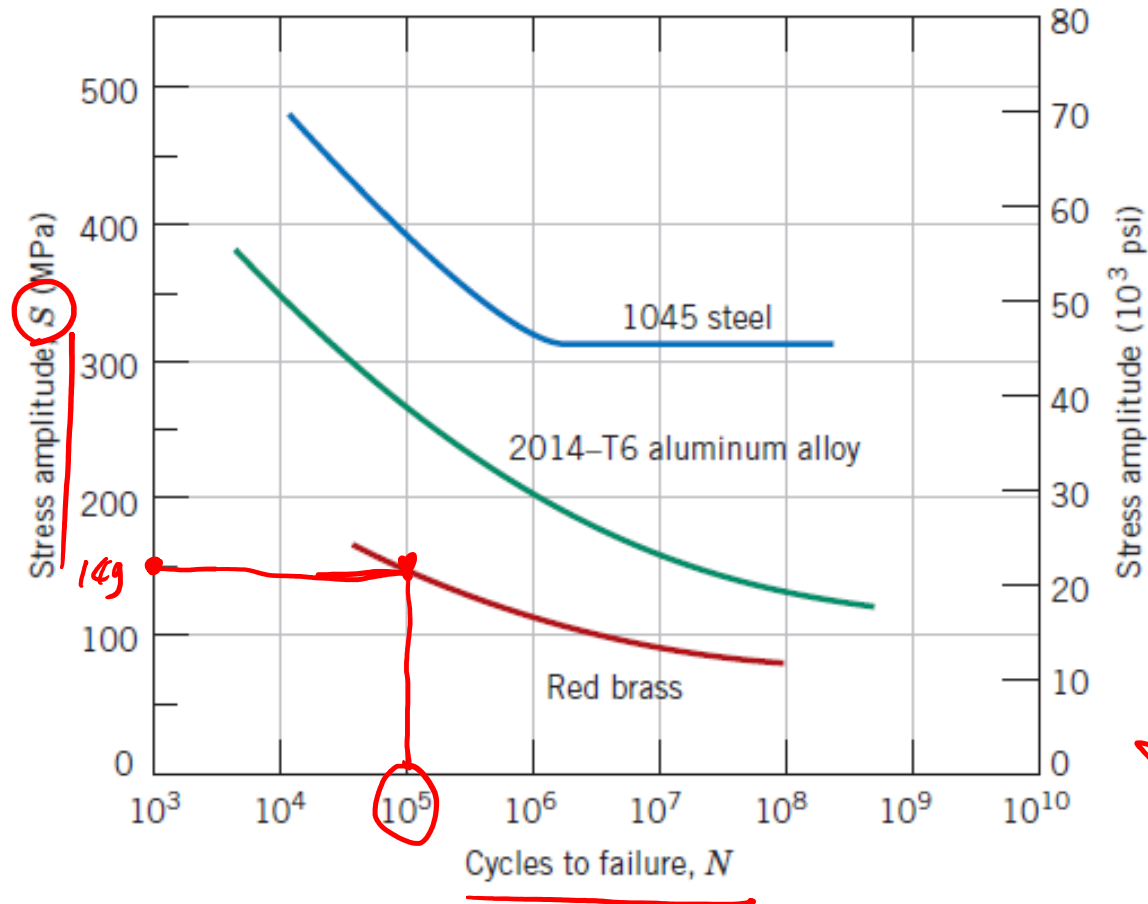
5- A Brass cylinder (8 mm diameter) is cyclically loaded at +7500 N e -7500 N along its axis. Use curves below to calculate its fatigue life.



5- A Brass cylinder (8 mm diameter) is cyclically loaded at +7500 N e -7500 N along its axis. Use curves below to calculate its fatigue life.



$N = ?$



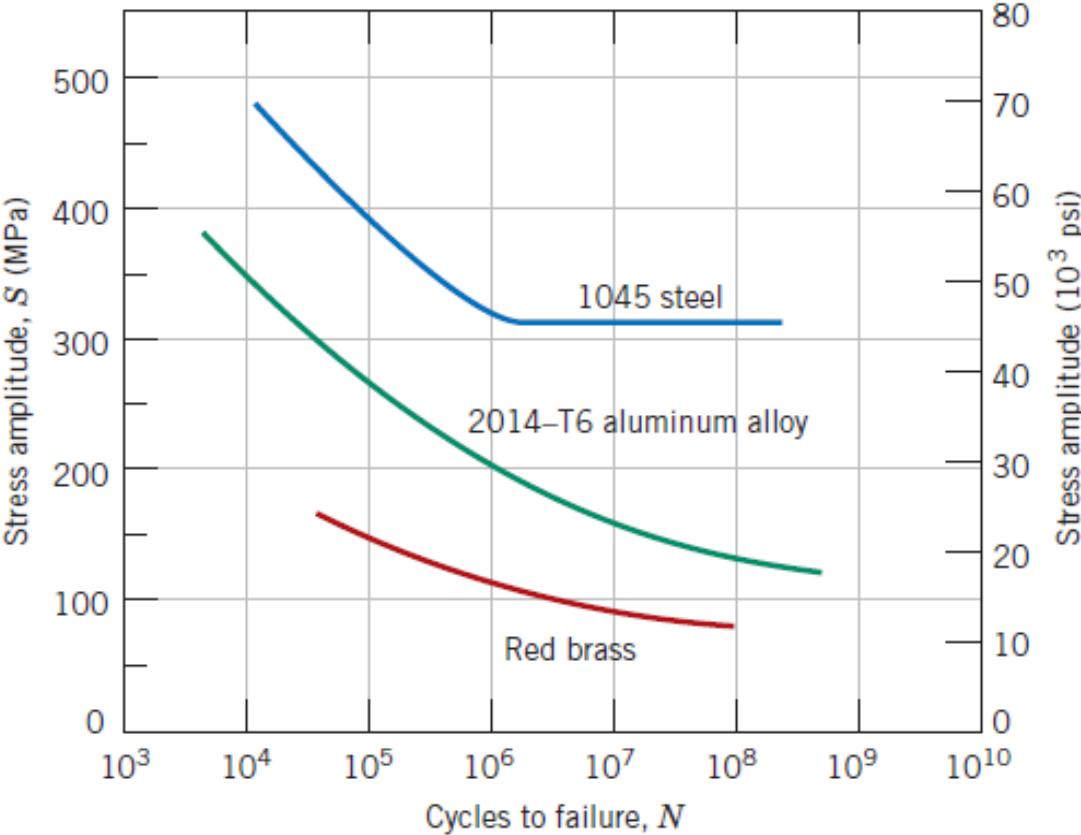
$$A = \pi \cdot r^2 = (4 \text{ mm})^2 \cdot \pi = 50,3 \text{ mm}^2$$

$$\sigma = \frac{F}{A} = \frac{7500 \text{ N}}{50,3 \text{ mm}^2} = 149 \text{ MPa}$$

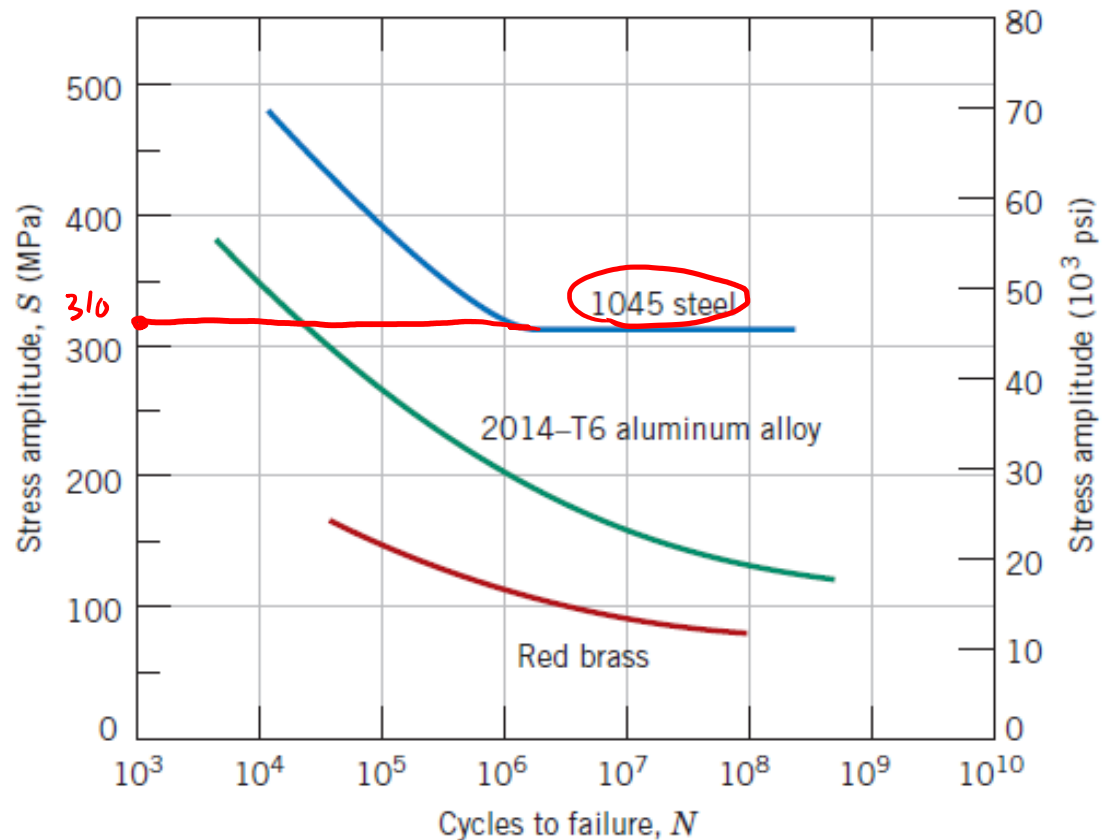
$$N = 10^5 \text{ cycles}$$

A

A steel rod is cyclically loaded at $S= 22 \text{ kN}$ along its axis. Use curves above to calculate the minimum diameter for this component to avoid fatigue fracture. Use a safety coefficient 2.



A steel rod is cyclically loaded at $S = 22 \text{ kN}$ along its axis. Use curves above to calculate the minimum diameter for this component to avoid fatigue fracture. Use a safety coefficient 2.



$$S < S_{FL} \quad S_{FL} = 310 \text{ MPa}$$

$$S = \frac{S_{FL}}{2} = \frac{310}{2} = 155 \text{ MPa}$$

$$\frac{[N]}{[N/mm^2]} = mm^2$$

$$S = \frac{F}{A} \quad A = \frac{F}{S} = \frac{22000 \text{ N}}{155 \text{ MPa}} = 141,9 \text{ mm}^2$$

$$A = \pi \cdot r^2 \quad r = \sqrt{\frac{A}{\pi}} = 6,72 \text{ mm}$$

$$d = 13,44 \text{ mm}$$

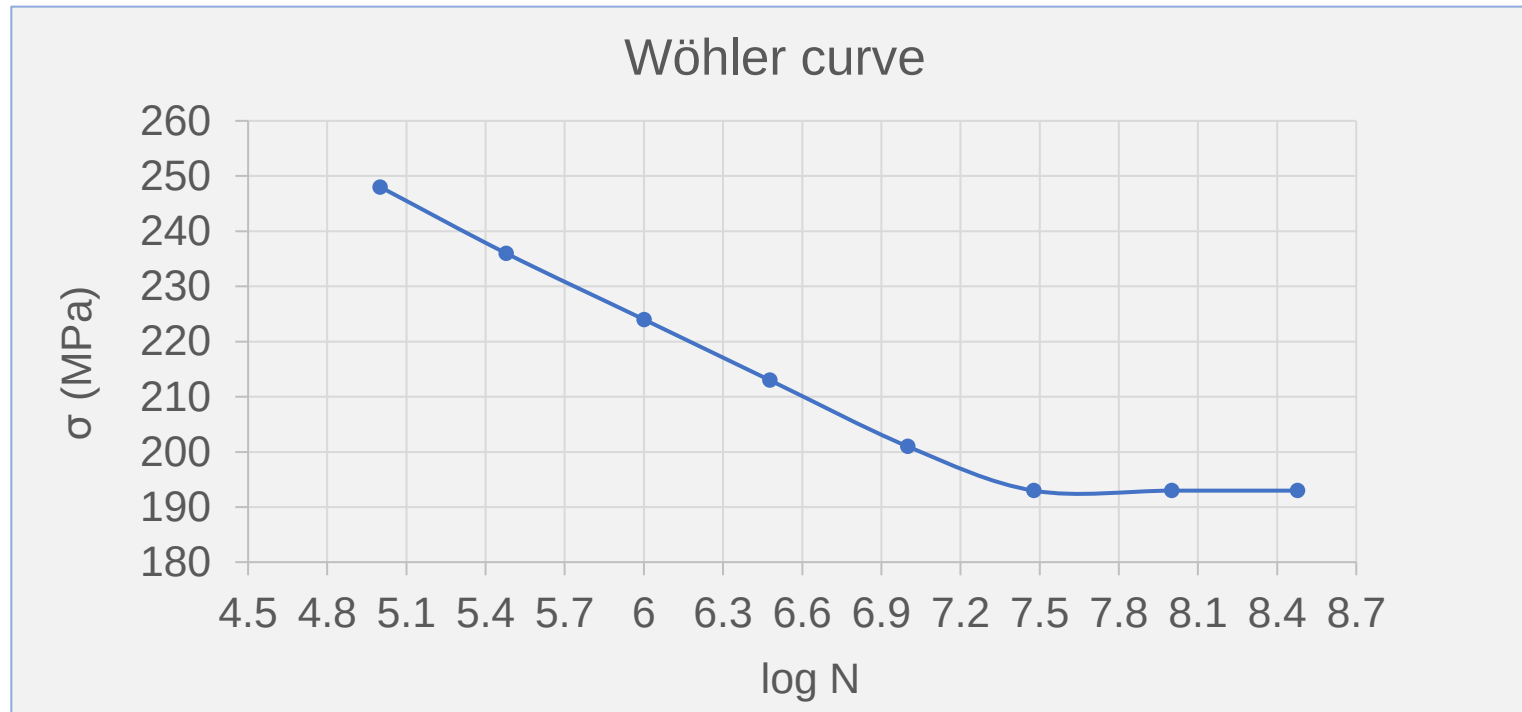
Es 5. The data obtained from a series of fatigue tests for a material are as follows:



σ (MPa)	N°
248	$1 \cdot 10^5$
236	$3 \cdot 10^5$
224	$1 \cdot 10^6$
213	$3 \cdot 10^6$
201	$1 \cdot 10^7$
193	$3 \cdot 10^7$
193	$1 \cdot 10^8$
193	$3 \cdot 10^8$

After constructing the Wöhler curve, determine:

- a) The fatigue limit
- b) The number of cycles to failure at 230 and 175 MPa
- c) The fatigue strength at $2 \cdot 10^5$ and $6 \cdot 10^6$ cycles



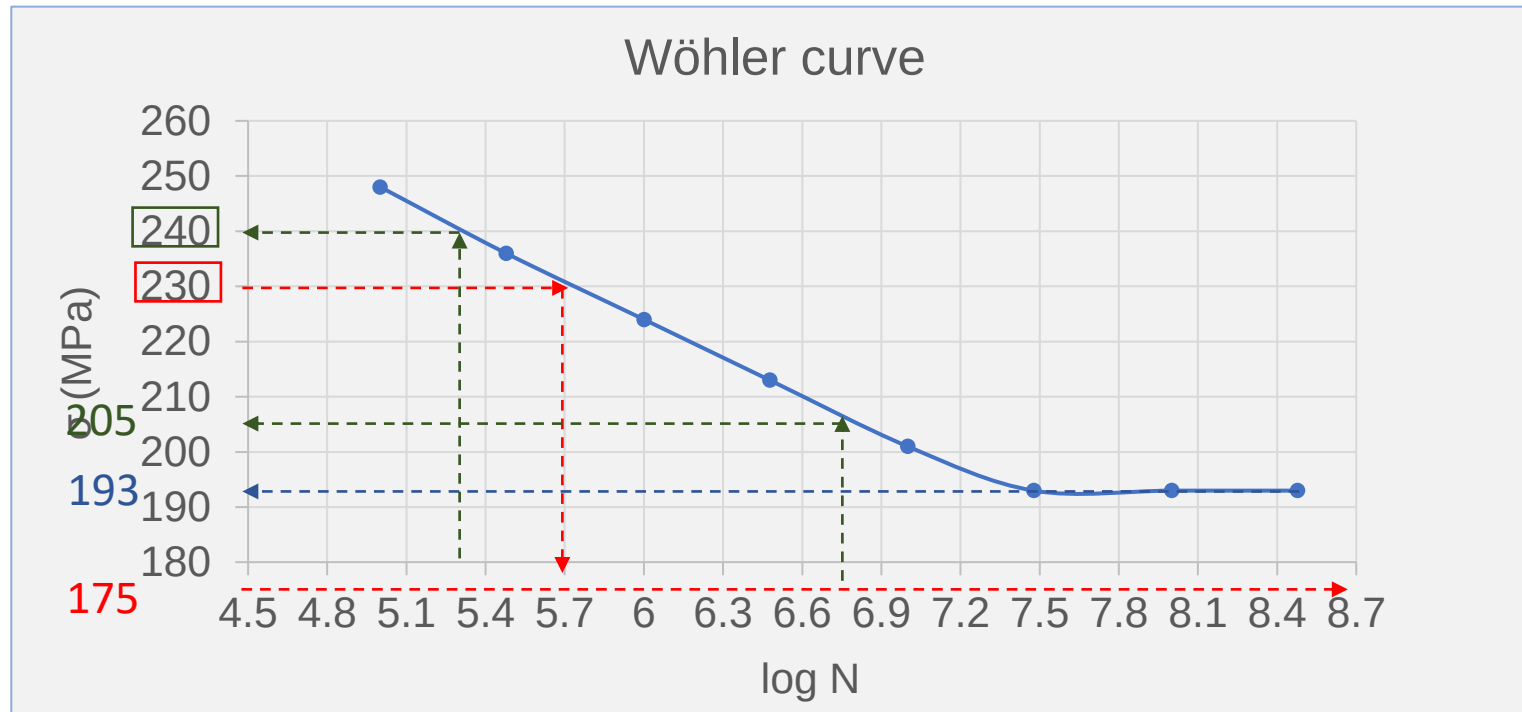
a) Fatigue limit :

b) Number of cycles to failure at 230 MPa:

Number of cycles to failure at 175 MPa:

c) Fatigue strength at $2 \cdot 10^5$ cycles:

Fatigue strength at $6 \cdot 10^6$ cycles:



a) Fatigue limit : 193 MPa

b) Number of cycles to failure at 230 MPa: 501 187; $N = 5 \cdot 10^5$

Number of cycles to failure at 175 MPa: ∞

c) Fatigue strength at $2 \cdot 10^5$ cycles: $\log (2 \cdot 10^5) = 5.3 \rightarrow 240 \text{ MPa}$
 Fatigue strength at $6 \cdot 10^6$ cycles: $\log (6 \cdot 10^6) = 6.8 \rightarrow 205 \text{ MPa}$

6- Draw the fatigue curve (S/N) of a metal having the following surface characteristics: polished surface; sand-blasted surface; U-notches surface; V-notched surface. Explain differences.

1- Polished



2- Sand-blasted



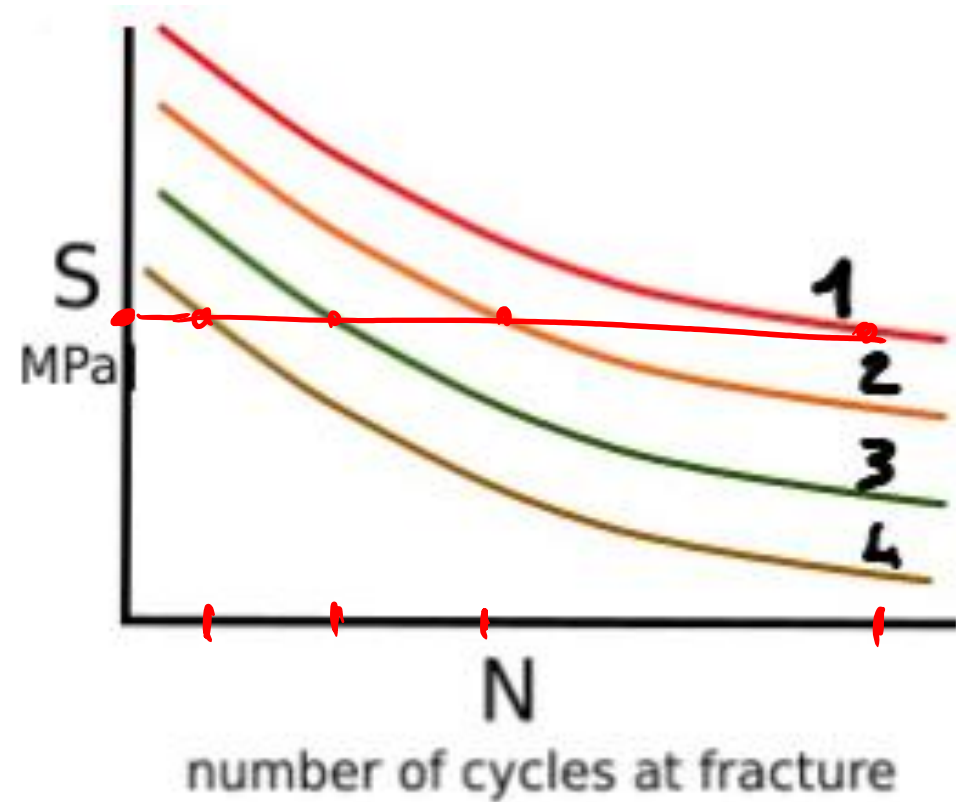
3- U-notch



4- V-notch



6- Draw the fatigue curve (S/N) of a metal having the following surface characteristics: polished surface; sand-blasted surface; U-notches surface; V-notches surface. Explain differences.



1- Polished



2- Sand-blasted



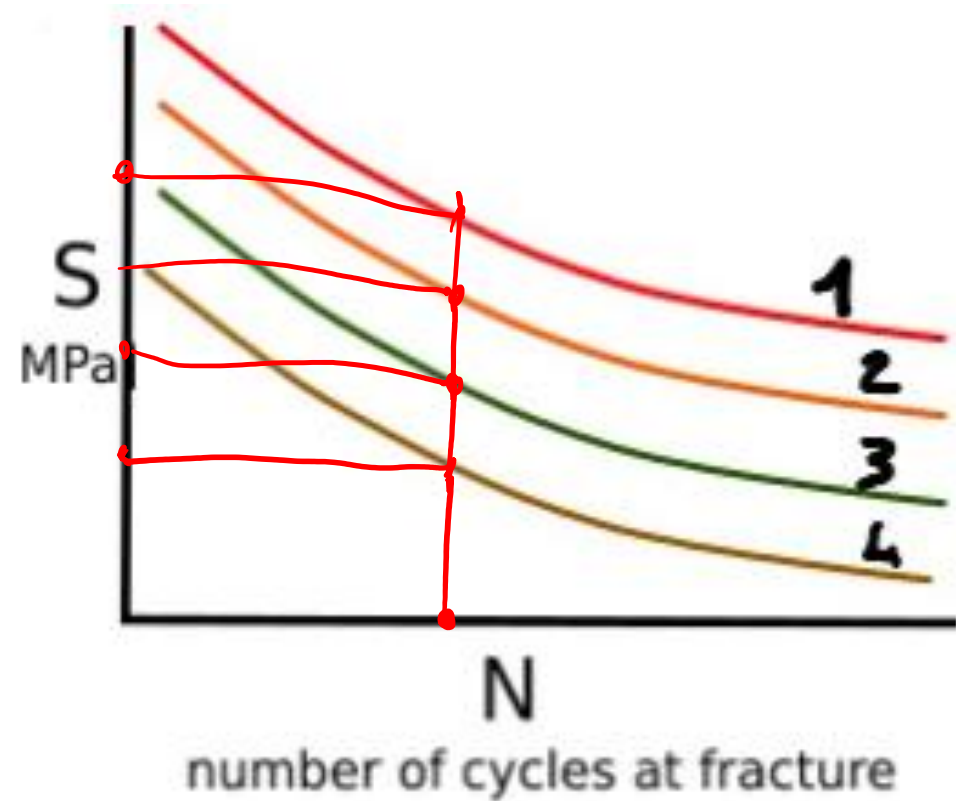
3- U-notch



4- V-notch



6- Draw the fatigue curve (S/N) of a metal having the following surface characteristics: polished surface; sand-blasted surface; U-notches surface; V-notched surface. Explain differences.



1- Polished



2- Sand-blasted



3- U-notch



4- V-notch

